Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of in-house industry experience at HDFC Asset Management building and running production GenAI applications end to end: tool-calling agents and RAG pipelines over PostgreSQL with pgvector, FastAPI services and a ReactJS frontend, with LDAP-based access control, deployed in Docker through CI/CD. Took each use case from design through release and iteration with Compliance, Client Service and Investment stakeholders, with Compliance reviewing every application before release; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Recently completed an MSc in Data Science at the University of Bristol (result awaited), with a dissertation in partnership with AstraZeneca. Seeking the AI Engineer role at Norton Rose Fulbright, building agentic workflows, grounded retrieval and the interfaces around them for legal services.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Agentic AI, RAG & LLM Workflows in Production

- Designed and built **RAG** pipelines and knowledge retrieval for internal **GenAI** applications used across Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**), with a separate **pgvector** database and **Azure OpenAI** embeddings for each department so access stayed siloed and similar content from different departments was not confused; built end to end with a **ReactJS** frontend and **Python FastAPI** microservices.
- Built **agentic tool-calling** workflows where AI agents call **Python functions** as tools to run SQL-backed fund-comparison calculations and generate **PDF** comparison reports on user request; guardrailed responses through **system prompts** and versioned every prompt change in **Git**.
- Built a **Python** email-processing automation that turns over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy.
- Released each application to real users, iterated on what broke, and had the **Compliance** department review every application before release. Chose models by cost: **Llama-3 self-hosted on a GPU EC2 instance** for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.

APIs, Data Integration & User Interfaces

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); integrated it with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from **Excel** to Power BI and improving analyst efficiency by **10 hours a week**; recreated them in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.

Secure Deployment, Access Control & Team Delivery

- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** utilizing core services such as **EC2, S3, ECR** and **Lambda**, with **Docker** microservices (FastAPI, uvicorn), separate frontend and backend repositories and containers, and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools that replace manual work; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: JSON requested and validated with Pydantic, malformed output repaired, partial results salvaged, bad results never cached, vision calls retried once. A 40 RPM limit shaped the design: 12 calls instead of 108, calls serialised with a semaphore, results streamed to the UI over server-sent events and cached by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and shows the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), TypeScript (React frontend, CellLineSelector), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite